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Title: Early Prediction of Customer Escalation in Support Tickets Using Context-Aware Retrieval-Augmented Analytics

Abstract -

Escalation of software support tickets can create serious pressure for support teams and can also increase operational costs for software companies. When a ticket escalates, it often requires more staff time and sometimes involvement from higher-level support engineers or management. For this reason, identifying potential escalation earlier in the process could help teams respond more effectively and improve the overall customer experience. A large portion of previous research has focused on sentiment analysis or text classification techniques. Sometime these approaches are useful, they often detect escalation only after the customer frustration becomes explicit.

This study looks at whether it is possible to predict ticket escalation by learning from historical records. In many support environments, different tickets tend to follow similar patterns, even if the technical issues are not identical. Recognising these similarities may help identify cases that are more likely to escalate later. Another aspect of the study involves examining how customer sentiment changes as the support interaction progresses.

Traditional sentiment analysis methods do not always capture these early signals. The primary reason is typically analysing messages individually rather than looking at the conversation as a continuous interaction. When messages are treated separately, the broader context of how the discussion evolves over time can be missed. To address this issue, this study proposes a context-aware analytics framework that combines conventional machine learning models with information retrieved from previously resolved support tickets. Through the use of embedding-based similarity search, the system can locate past tickets that share certain characteristics with the current case. This allows the model to incorporate additional context, such as how similar tickets were resolved, the level of interaction between customers and support staff, and whether those cases eventually escalated.

Current study will compare baseline machine learning models with versions that include these additional retrieval-based contextual features. Performance will be evaluated using commonly used evaluation metrics for classification.

Research Questions:

RQ1: Can retrieval-augmented analytics help improve the early prediction of support ticket escalation compared to using traditional machine learning models?

RQ2: Does modelling sentiment progression across ticket conversations improve escalation prediction accuracy?

Datasets:

1. https://huggingface.co/datasets/rtweera/customer_care_emails
2. https://www.kaggle.com/datasets/ajverse/customer-support-tickets-crm-dataset
3. https://huggingface.co/datasets/KameronB/synthetic-it-callcenter-tickets